D. Palindrome Flipping

time limit per test: 2 seconds
memory limit per test: 256 megabytes

You are given two binary strings s and t, each of the same length n. You are allowed to perform the following operation:

- Pick indices l, r ($1 \le l < r \le n$) such that substring $s_{l,r}$ is a palindrome and flip all bits in substring $s_{l,r}$.

The goal is to finally make s equal to t, performing any of the above operations at most $2n$ times (possibly none).

A substring $s_{l,r}$ of a string s is the contiguous sequence of characters starting from index l and ending at index r (both inclusive), where $1 \le l < r \le |s|$. Here $|s|$ denotes the length of the string s.

A string is a palindrome if it reads the same forwards and backwards. For example, the strings `101` and `00` are palindromes, while `10` is not.

Flipping all bits in a substring means changing each `0` to `1` and each `1` to `0` in that substring. For example, flipping the substring `101` results in `010`.

Input

Each test contains multiple test cases. The first line contains the number of test cases T ($1 \le T \le 5 \cdot 10^3$). The description of the test cases follows.

The first line of each test case contains an integer n ($4 \le n \le 100$) — the length of the strings s and t.

The next two lines of each test case contain the binary strings s and t, respectively.

It is guaranteed that the sum of n^2 over all test cases does not exceed $5 \cdot 10^5$.

Output

For each test case, if it is impossible to achieve the goal, print -1.

Otherwise, the first line should contain an integer k ($0 \le k \le 2n$) — the number of operations.

For each of the next k lines, print two integers l, r ($1 \le l < r \le n$) — the indices you choose in each operation. Note that $s_{l,r}$ must be a palindrome at this stage.

Example

input Copy

```
3
5
01011
10000
7
1010101
0101010
4
0010
0010
```

output Copy

```
2
1 3
3 5
```


→ Score table

	Score
Problem A	348
Problem B	870
Problem C	1044
Problem D	1392
Problem E	1914
Problem F1	1740
Problem F2	696
Successful hack	100
Unsuccessful hack	-50
Unsuccessful submission	-50
Resubmission	-50

* If you solve problem on 01:16 from the first attempt

```
1
1 7
0
```

Note

For the first test case:

- Initially, $s = \texttt{01011}$ and $t = \texttt{10000}$.
- First, we choose $l = 1$ and $r = 3$. This is a valid operation since $s_{1,3} = \texttt{010}$ is a palindrome. After flipping $s_{1,3}$, $s = \texttt{10111}$.
- Then, we choose $l = 3$ and $r = 5$. This is a valid operation since $s_{3,5} = \texttt{111}$ is a palindrome. After flipping $s_{3,5}$, $s = \texttt{10000}$.
- Now, s is equal to t.

For the second test case:

- Initially, $s = \texttt{1010101}$ and $t = \texttt{0101010}$.
- First, we choose $l = 1$ and $r = 7$. This is a valid operation since $s_{1,7} = \texttt{1010101}$ is a palindrome. After flipping $s_{1,7}$, $s = \texttt{0101010}$.
- Now, s is equal to t.

For the third test case:

- Initially, s is equal to t. No operations are needed.